

Estimating Throughput for Urban Air Mobility Operations

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Urban Air Mobility (UAM) is a new air transportation service concept to carry passengers or cargo in urban areas with advanced technologies. To evolve the UAM ecosystem towards the higher level of operational density and complexity, appropriate airspace construct design and airspace management services should be developed. For efficient air traffic management and strategic airspace resource planning for UAM operations, it is necessary to estimate the capacity of airspace resources and the maximum throughput that can be achieved by the given airspace construct design. This paper proposes a simple analytical method to estimate the maximum throughput for a given route structure by combining existing capacity estimation approaches for individual resources like vertiports and fleet availability and considering other factors that can affect the throughput. The proposed throughput estimation model is applied to a simple UAM route network in Dallas/Fort Worth metropolitan area and evaluated with a realistic traffic scenario. The evaluation results show that this throughput estimation model can identify the most significant factor affecting the capacity and predict how the throughput changes in dynamic operating environments.

I. Nomenclature

$Capacity_{Network}$	=	Total network capacity for a given UAM route network (operations/hour)
C_{cross}	=	Capacity at a crossing waypoint for UAM operations (operations/hour)
C_{fleet}	=	Fleet availability capacity for UAM operations (operations/hour)
C_{pads}	=	Vertipad capacity of a vertiport (operations/hour)
C_{route}	=	In-trail route capacity for UAM operations (operations/hour)
C_{sector}	=	Airspace sector capacity (operations/hour)
$C_{surface}$	=	Surface capacity of a vertiport (operations/hour)
C_{vpt}	=	Vertiport capacity (operations/hour)
$C(t)$	=	Total capacity at time t
$D(t)$	=	Total demand at time t
FT_j	=	Flight time along route j (minutes)
N_f	=	Number of UAM vehicles that operators in an UAM ecosystem can provide

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N_{ops}	=	Maximum number of concurrent airborne operations in a given UAM route network
N_p	=	Number of vertipads available at a vertiport
N_s	=	Number of parking spaces available at a vertiport
R	=	A set of all routes operated in the given UAM route network
R'	=	A set of exclusive routes in the given UAM route network
S	=	A set of airspace sectors accommodating the given UAM route network
S_{sep}	=	Required spatial separation between flights
$T_{available}$	=	Net time that a UAM vehicle is available for UAM operations (hours)
$T_{operation}$	=	Average end-to-end time to complete an operation by a single UAM vehicle (hours)
T_{sep}	=	Required temporal separation between flights (minutes)
t_{arr}	=	Amount of time reserved on a vertipad for arrivals (minutes)
t_{dep}	=	Amount of time reserved on a vertipad for departures (minutes)
$t_{surface}$	=	Aircraft movement time on the surface for taxiing (minutes)
t_{window}	=	Desired time window for vertiport usage (minutes)
V	=	A set of active vertiports located within the given UAM network
v	=	Cruise speed of UAM aircraft
θ	=	Crossing angle between two intersecting routes (degrees)

II. Introduction

Urban Air Mobility (UAM) is an emerging air transportation service concept that carries passengers or cargo in and around urban areas using quiet and efficient vehicles [1, 2]. UAM is expected to reduce the door-to-door travel times and their uncertainty significantly, compared to the ground transportation in congested metropolitan areas. Many vehicle manufacturers in the world are developing electric vertical takeoff and landing (eVTOL) aircraft in diverse configurations and performances for UAM operations and conducting flight tests of their prototype models for commercialization.

New airspace management concepts need to be developed for these new types of aircraft because the current air traffic management system resources cannot accommodate UAM operations that will scale up over time. The Federal Aviation Administration (FAA) proposed the UAM Corridor concept for its UAM ConOps 1.0 operations [3]. The UAM Corridors are a performance-based airspace of defined dimensions in which aircraft abide by UAM specific rules, procedures, and performance requirements. Both UAM and non-UAM aircraft can fly within or cross the corridors, as long as they meet the performance and participation requirements. However, it is not clear how this concept can accommodate a high operational tempo as the route network structure, regulations, airspace procedures, and aircraft automation level mature. From the Unmanned Aircraft System (UAS) Traffic Management (UTM) paradigm, the Provider of Services for UAM (PSU) is also introduced for efficient data exchanges and information sharing between multiple UAM operators. With this new concept and system architecture for a UAM ecosystem, more research is needed for both airspace design and airspace management services.

NASA and industry partners have been conducting a series of simulations to develop a UAM airspace roadmap, test airspace management services, define requirements for a UAM airspace management system, and address foundational issues for matured UAM operations [4-8]. In the first simulation activity, the information exchange requirements for initial UAM operations were explored using the existing helicopter routes in Dallas/Fort Worth (DFW) airspace. The next simulation evaluated the operational volumes for two UAM operators to share operational information on a set of routes connecting large airports in the DFW area using the UTM architecture. Through the following simulation, PSUs developed and provided by NASA and industry partners were connected and tested. The latest simulation activity, called X4, focused on the Strategic Conflict Management (SCM) services for UAM operations. As an initial SCM approach, Demand Capacity Balancing (DCB) capabilities, including demand-capacity imbalance detection and resolution between multiple operators at vertiports, were examined in various traffic scenarios [8-12]. Given the static vertiport capacity, the throughput in the pre-defined route structure and the pre-departure delay to resolve the demand-capacity imbalances were measured as major evaluation metrics. These DCB capabilities may be expanded to consider other airspace resources like crossing and merging waypoints along the routes and dynamic operational environment such as capacity reduction due to adverse weather.

For efficient air traffic management and strategic resource planning and investment, it is important to correctly estimate the airspace resource capacity and the maximum throughput that can be achieved by the given airspace design, procedures, and resources. As seen in the aforementioned X4 simulation, the throughput is used as one of the main metrics to evaluate the performance of airspace management services.

In traditional air traffic management, capacity estimation was studied mainly for airports as the main bottleneck to restrict throughput. Out of several airfield elements such as runway, taxiway, and apron (ramp area), the runway capacity is usually critical because most delays occur at runways. The main factors affecting the capacity of a runway system include the number and geometric layout of the runways, ATM separation requirements, visibility, ceiling, precipitation, wind direction and strength, aircraft fleet mix, mix and sequencing of movements (arrivals versus departures), type and location of runway exits, state and performance of the ATM system, and noise considerations [13].

However, UAM operations are different from traditional airline flight operations in the National Airspace System (NAS) with respect to various aspects, making it difficult to apply the same approach to estimating capacity and throughput. First, the expected flight time of UAM operations is much shorter. Also, it is expected that the tactical separation services for the flights operated within the UAM Corridors are not provided by air traffic control (ATC). Lastly, UAM flights are operated within the given route network in a limited area around a city. Therefore, it is necessary to look at the capacity of the UAM route network and estimate its achievable throughput, considering the characteristics of the UAM ecosystem. Furthermore, as UAM evolves with advancements of the relevant technologies like autonomous flight toward UAM Maturity Level (UML)-4 corresponding to higher operational tempo [14, 15], it is important to estimate how many flights can operate in the given UAM network for efficient and safe flight planning and operations. In addition, this throughput estimate would be beneficial for informing timely investment on the UAM aircraft and the relevant infrastructure construction.

In this paper, therefore, we propose an analytical method to estimate the maximum throughput for a given airspace structure by combining existing capacity estimation approaches for individual resources like vertiports and fleet availability. This method can be used to evaluate the impact of various factors affecting capacity and throughput and identify the most significant factor in different operating environments. It is expected that this work can help improve the airspace design and procedures for the accessibility and scalability of UAM operations.

In the following section, we define the throughput for UAM operations and discuss several factors affecting the throughput. Section IV describes how to estimate the capacity of individual resources in the UAM ecosystem. In Section V, an analytical model for throughput estimation for a given UAM network is proposed with some examples. The evaluation of the proposed method with various use cases will be provided in Section VI. Then, this paper concludes with contributions and a future work plan in Section VII.

III. Throughput for UAM Operations

In this section, we define the throughput for UAM operations and discuss several factors affecting the throughput.

A. Throughput definition

The throughput for UAM operations can be defined as the observed number of operations on a resource over a given time period. Considering the relatively short flight time of UAM operations with an expected range from 5min to 20min around an urban area, the time period can be as short as a few minutes, but we will use a typical time period of 1 hour in this paper to capture a substantial number of operations. This definition can be applied to various airspace resources such as vertiport, air route, crossing or merging waypoint, and airspace sector. At vertiports, the throughput applies to both aircraft departures and arrivals.

Another metric to measure the traffic volume in a given UAM network is the number of simultaneous operations in the air. This metric keeps changing over time, depending on takeoff and landing operations. The maximum number of simultaneous operations can be used to estimate the throughput achievable in the given route structure, while keeping the separation minima between flights for safety.

B. Factors affecting throughput

There are many factors that can affect the throughput of UAM traffic in the UAM ecosystem. First, the airspace construct which includes the vertiport locations, the layout of UAM corridors, and the number of vertiports and routes connecting those vertiports determines the rough size of air traffic volume with which UAM service providers can operate. Obviously, the greater the numbers of vertiports and routes, the higher the maximum throughput. Second, airspace resource capacity is a main bottleneck to limit the maximum throughput, which includes vertiport capacity, air route capacity, and airspace sector capacity. The resource availability from UAM operators is another important factor to define the throughput. If the operators cannot provide a sufficient fleet with pilots and ground crew to meet the increasing trip requests appropriately, the growth of the throughput can be limited by the operator resource availability. If the resources of airspace and operator are sufficient or the traffic demand is low (UML-1 or -2), the traffic demand level may determine the actual throughput. Air traffic management also needs to support the emerging

operations of UAM properly. The flight rules applied like Visual Flight Rules (VFR) and Instrument Flight Rules (IFR), the workload of air traffic controllers who monitor and manage the interactions between UAM flights and other air traffic in the controlled airspace, and the minimum separation requirements for safe operations can affect the maximum throughput capacity. Lastly, the weather conditions like wind, gust from buildings, visibility, precipitation, and noise restrictions can impact the throughput. These factors affecting the throughput of UAM operations are listed in Table 1.

Table 1: Factors affecting throughput of UAM traffic.

Category	Factors
Airspace construct	Number of available routes, number of vertiports, no-fly zones
Traffic demand	Number of departures and arrivals
Airspace resource capacity	Vertiport capacity, route capacity, sector capacity
Operator resources	Pilots (onboard or remote), aircraft, ground crew, etc.
Air traffic management	Flight rules (VFR vs. IFR), air traffic controller workload, minimum separation between aircraft (in trail, crossing, merging, etc.)
Others	Weather (wind, visibility, precipitation, etc.), noise

According to Ref. [16], key operational constraints that may limit the growth of UAM systems include the scalability of the existing air traffic control (ATC) system, the availability of aviation ground infrastructure, and the noise acceptability to the public. Among the ATC scalability constraints, reduced separation, increased ATC automation, and trajectory-based operations can increase the capacity, whereas heterogeneous aircraft, large procedure containments, inclement weather, and frequent Temporary Flight Restrictions (TFRs) may reduce the capacity. Regarding the infrastructure availability, the increased number of vertipads, gates, and aircraft staging areas, reduced aircraft turnaround time, and homogenous fleet mix can enhance practical capacity, whereas noise-based operational restrictions, IFR, and obstructed procedures in urban landscape may reduce the capacity and accessibility. These elements in Ref. [16] roughly coincide with the factors listed in Table 1. In this paper, we assume that the airspace construct that requires long-term investment is pre-determined and try to formulate the capacity estimation of some major factors affecting the traffic density and throughput of UAM operations in the given airspace route structure.

IV. Capacity Estimation

Capacity is a main factor to determine the maximum throughput. In general, capacity of airspace resources for UAM operations depends on airspace procedure and design, infrastructure, requirements, and Community Based Rules (CBRs) [17, 18]. It can also change in a short time, according to the operational environment such as weather conditions. For the capacity, various airspace resources should be considered, including vertiport, route, constrained waypoints (e.g., crossing, merging, and spacing), and airspace sector.

A. Vertiport capacity

In traditional air traffic management, airport capacity is considered a main bottleneck to limit the growth of air traffic volume. Similarly, vertiports would be the most significant element to determine the maximum capacity of the airspace resources in the UAM ecosystem. There are many factors affecting vertiport capacity, which include the number of vertipads for takeoff and landing operations, the number of parking spaces and passenger/cargo loading spaces, the number of battery charging stations for eVTOL aircraft, the layout of a vertiport like locations of vertipads and parking area [19], the size and design of surface movement area for taxiing, and weather condition such as wind, precipitation, and visibility.

Guerreiro et al. formulated the vertiport capacity estimation method, considering the number of vertipads and parking spaces [20]. First, the surface capacity of a vertiport is given below:

$$C_{surface} = N_s \cdot \frac{t_{window}}{t_{surface}} \quad (1)$$

where N_s is the number of parking spaces available at a vertiport, t_{window} is a desired time window (one hour in this paper), and $t_{surface}$ is the movement time on the surface (for taxiing in non-vertipad area).

As the number of runways makes a direct impact on the airport capacity, the number of vertipads is one of the most important factors to determine the vertiport capacity. The vertipad capacity can be defined by Eq. (2):

$$C_{pads} = 2 \cdot N_p \cdot \frac{t_{window}}{t_{arr} + t_{dep}} \quad (2)$$

where N_p is the number of vertipads at a vertiport, and t_{arr} and t_{dep} are the amounts of time reserved on a vertipad for arrival and departure, respectively. Note that the average operation time of departure and arrival is taken in this equation, assuming that takeoff and landing operations are served at each vertipad evenly.

Then, the vertiport capacity (in operations per hour) can be obtained by the following:

$$C_{vpt} = \min\{2 \cdot C_{surface}, C_{pads}\} \quad (3)$$

In this equation, we multiply the surface capacity in Eq. (1) by 2 with the assumption that each surface operation is associated with two vertipad operations (an arrival and a departure). Since the theoretical vertipad capacity assumes independent operations to all vertipads, the actual vertiport capacity may be lower, depending on the vertipad layout and configuration. Also, the vertiport capacity can be affected by weather conditions and local restrictions (e.g., noise), which are not considered in this paper and remained for future work.

B. In-trail route capacity

For safe operations, two consecutive flights should maintain the minimum separation while flying along a given route. The hourly in-trail route capacity is determined by the minimum separation time T_{sep} (in minutes), which can be computed by dividing the minimum spatial separation by the cruise speed of the aircraft. Therefore, the in-trail route capacity (in operations per hour) can be expressed by:

$$C_{route} = \frac{60}{T_{sep}} = \frac{60 \cdot v}{S_{sep}} \quad (4)$$

where S_{sep} is the required spatial separation between flights, and v is the cruise speed of the aircraft.

In Eq. (4), it is assumed that there is no wind effect and that all the aircraft operate at the same cruise speed for simplicity. If those uncertainties are considered, the capacity value will be reduced. As the UAM concept of operations, procedures, and aircraft models mature, this assumption may be revisited to improve the capacity modeling. Also, the separation minima S_{sep} can be different, depending on the operational condition such as VMC and IMC. Under the IFR, the required minimum separation can increase, leading to less route capacity.

C. Crossing waypoint capacity

When two flights coming from each route meet at an intersection, those flights should be separated with each other properly to avoid a potential conflict. This separation requirement limits the number of flights that can go through the crossing waypoint. With the same assumptions made in Eq. (4), the capacity at a crossing waypoint (in operations per hour) can be defined as follows [21]:

$$C_{cross} = \frac{60}{T_{sep}} = \frac{60 \cdot v \cdot \cos(\frac{\theta}{2})}{S_{sep}} \quad (5)$$

where T_{sep} and S_{sep} are the required temporal and spatial separation at a crossing waypoint, respectively, v is the cruise speed of the aircraft, and θ is the crossing angle between two intersecting routes.

Note that the minimum temporal separation between crossing flights depends strongly on the inbound crossing angle θ , i.e., increases non-linearly as a function of crossing angle. In this equation, 0° and 180° crossing angles correspond to in-trail and head-on flight conditions, respectively. Eq. (5) is also valid at merging waypoints where two flights from different routes meet and start sharing the same route, since they require the minimum separation when entering the merging waypoint. Figure 1(a) illustrates the crossing waypoint capacity curves as a function of crossing angle with respect to the selected separation minima.

D. Sector capacity

If the airspace in which UAM flights are operated is controlled by air traffic controllers, the maximum number of UAM operations can be limited by the workload and other constraints by the controllers who are responsible for monitoring and controlling all the aircraft operating in the given sector (or any divided airspace), including UAM vehicles, helicopters, general aviation, and airline flights. This capacity (in operations per hour) is called the sector capacity, C_{sector} , in this paper.

E. Fleet availability

From the UAM operator side, the fleet availability can limit the maximum throughput of UAM operations, if neither the fleet size is enough to accommodate the demand nor the fleet is managed efficiently. Li et al. derived an upper bound for the maximum demand rate that a fleet size can fulfill [22], which we refer to as the fleet availability capacity. This capacity (in operations per hour) can be expressed as follows:

$$C_{fleet} = \frac{\frac{T_{available}}{\Delta t}}{T_{operation}} = \frac{N_f}{T_{operation}} \quad (6)$$

where $T_{available}$ represents the net time the fleet is available for UAM operations, $T_{operation}$ is the average end-to-end time required to complete an operation by a single vehicle in the fleet, and N_f is the total fleet size that operators in an UAM ecosystem can provide.

Clearly, the fleet availability capacity will increase when the fleet size, N_f , increases by adding more UAM vehicles. The single operation time, $T_{operation}$, includes the actual flight time from origin to destination and the turnaround time at a vertiport for a next departure that is spent for passenger/cargo loading/unloading, battery charging, maintenance, and so on. Hence, the capacity can also increase if either actual flight time or turnaround time is reduced by raising the cruise speed, improving the efficiency of the passenger boarding process, reducing battery charging time, and so forth. Figure 1(b) shows various fleet availability capacity values, corresponding to the different combinations of fleet size (N_f) and average operation time ($T_{operation}$).

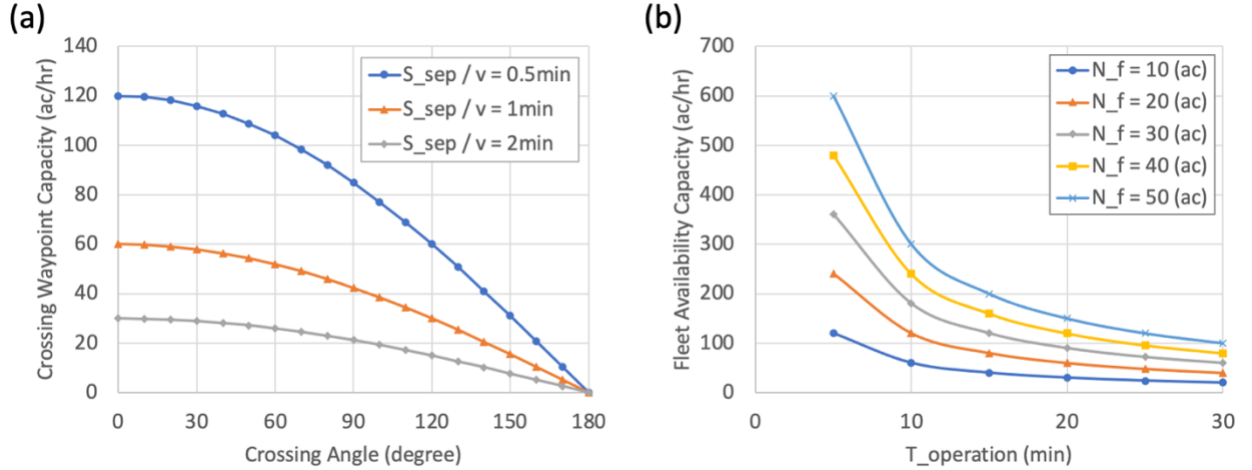


Fig. 1 (a) Crossing waypoint capacity as a function of crossing angle and separation minima, (b) Fleet availability capacity as a function of operation time and fleet size.

V. Analytical Model for Throughput Estimation

This section presents an analytical model to estimate the throughput of UAM operations in a given route network, with a few simple examples. It is noted that the proposed approach provides a rough throughput estimate with selected factors affecting the throughput and some assumptions like no wind effects and no uncertainties.

A. Theoretical throughput estimation

Typically, the throughput is determined by air traffic demand and capacity, although it can be affected by other factors. If the traffic demand exceeds the capacity, the throughput is the same as the given capacity, which is referred to as the maximum throughput capacity. In this case, large delay is expected until the imbalance between demand and capacity is resolved. On the contrary, if the capacity is sufficient to accommodate the traffic, its throughput is limited to the amount of demand, which can be referred to as the practical throughput. This situation doesn't cause any delay, but some resources may not be fully utilized. In summary, the throughput can generally be expressed by Eq. (7):

$$Throughput = \min\{Demand, Capacity\} \quad (7)$$

Note that the demand here represents the number of operations scheduled in a unit time. This traffic demand includes trip requests by customers, as well as repositioning flights with empty seats due to the spatial bias of the trip requests.

In the UAM ecosystem, it is naturally expected that traffic demand changes over time, showing some fluctuations with peaks and valleys on the demand profile. So, the demand $d(t)$ is a function of time t . In addition to the new trip requests at current time t , the total demand, $D(t)$, may include the spillover demand from the previous time $t-1$, if the demand exceeded the capacity at that time. Therefore, the demand for the realistic air traffic can be given like below:

$$D(t) = d(t) + \max\{D(t-1) - C(t-1), 0\} \quad (8)$$

where $D(t)$ and $C(t)$ are the total demand and capacity at time t , and $d(t)$ is newly generated demand at time t .

This demand model can be used to estimate the practical throughput that may change over time. Note that the capacity can also change in a short-term time period for various reasons such as weather condition change, temporary flight restrictions, and emergency landing.

B. Number of simultaneous operations

As mentioned in Section III.A, the number of simultaneous operations is another important metric to measure the current traffic congestion level and determine the maximum traffic level that the given UAM network can handle. This metric is also one of the criteria for evaluating the maturity level of UAM industry development [15]. For example, UML-3, which represents the proliferation of novel regulatory and airspace constructs and the supporting systems and services, expects to have up to 100 simultaneous airborne UAM operations in a metropolitan area, and UML-4 representing the integration of UAM operations into the NAS under more complex conditions will have hundreds (100-500) of concurrent UAM flights in the air.

Given the UAM route network, the maximum number of concurrent airborne operations, N_{ops} , can be estimated using the following equation:

$$N_{ops} = \sum_j \frac{FT_j}{T_{sep}}, \forall j \in R' \quad (9)$$

where FT_j represent the flight time along route j , T_{sep} is the required temporal separation between flights flown along the same route, and R' is a set of exclusive routes in the given network.

In a typical route network, most routes intersect with each other for the point-to-point shortest path, although some routes are independent. In Eq. (9), we introduce "exclusive routes" representing the routes that do not intersect with any others. When two or more routes are crossing, merging, or sharing a route segment, all these routes cannot be saturated by flights. In this case, it would be better to choose the longest route involved and evenly distribute flights by the minimum separation rather than to let the flights from different routes pass by the intersection point alternately. This strategy can both maximize the throughput and the number of simultaneous operations. For this reason, it is required to select the exclusive routes which have the longest flight time among the routes sharing the same crossing waypoint and are independent with other routes in the given route network.

C. Throughput for a given network

The throughput for a given UAM network can be determined by a combination of traffic demand and network capacity. Assuming that the traffic demand is high enough, however, the maximum throughput is determined by the total capacity of the network. Considering all the types of capacities mentioned above, the network capacity can be expressed as follows:

$$Capacity_{Network} = \min\{\sum_i C_{vpt,i}/2, \sum_j C_{route,j}, \sum_k C_{cross,k}, \sum_s C_{sector,s}, C_{fleet}\}, \forall i \in V, j \in R, k \in R, s \in S \quad (10)$$

where V is a set of active vertiports located within the given UAM network, R is a set of all routes operated, and S is a set of airspace sectors accommodating the given network.

For better understanding, here are two simple examples to calculate the maximum throughput and the number of simultaneous operations for a given network.

First, suppose that there are three one-way routes consisting of a triangular network, as shown in Fig. 2(a). We assume that the flight time of each route and the minimum separation time are 10 and 2 minutes, respectively, C_{vpt} for all vertiports is 60 operations/hour, N_f is 20 aircraft, $T_{operation}$ is 15 min (10 min of flight time and 5 min for turnaround time), and C_{cross} and C_{sector} are a large number M . Then, the capacity of this triangular network is computed as follows:

$$Capacity_{Network} = \min \left\{ 60 * \frac{3}{2}, \frac{60}{2} * 3, M, M, \frac{20}{15/60} \right\} = \min \{ 90, 90, M, M, 80 \} = 80 (ops/hr) \quad (11)$$

In this example, the factor determining the network capacity is the fleet capacity. If the operator adds 5 more aircraft, the fleet capacity can increase up to 100, and the network capacity will be 90 operations/hour. Also, assuming that the demand is always sufficient ($Demand=M$), the maximum number of simultaneous operations achievable in this network can be easily calculated as $N_{ops} = 10/2 * 3 = 15$ aircraft.

The second example in Fig. 2(b) considers an intersection of two routes, having the same flight time of 8 min and a perpendicular crossing angle ($\theta = 90^\circ$). Since this case has a crossing waypoint, the route capacity is replaced with the crossing waypoint capacity. With the same assumptions as the first case, the capacity of this network is calculated as follows:

$$Capacity_{Network} = \min \left\{ 60 * \frac{4}{2}, \frac{60}{2} * 2, \frac{60}{2 * 1.414} * 2, M, \frac{20}{13/60} \right\} = \min \{ 120, 60, 42.4, M, 92.3 \} = 42.4 (ops/hr) \quad (12)$$

Assuming that the demand is high enough, the maximum throughput for this network is therefore determined by the crossing waypoint capacity. The maximum number of simultaneous operations in this example is computed as $N_{ops} = 8 / 2 * 1 = 4$ aircraft since the number of exclusive routes is only one.

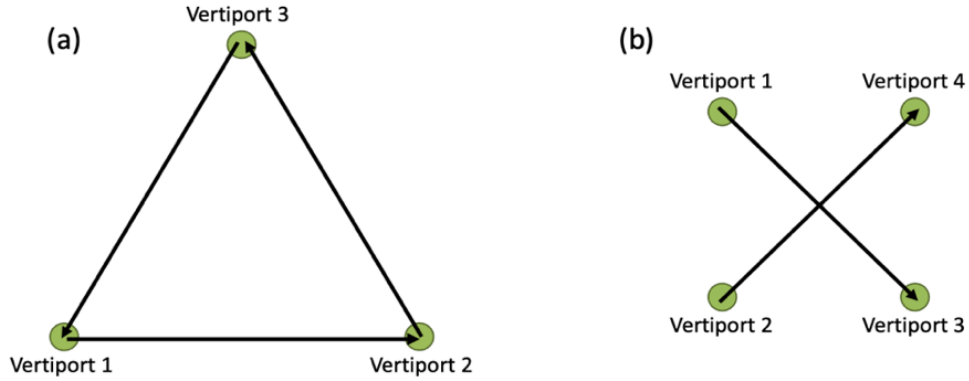


Fig. 2 Two route network examples for capacity estimates. (a) Triangular routes, (b) Crossing routes.

VI. Evaluation of a Throughput Estimation Model

The proposed throughput estimation model can be applied to different UAM networks having more complicated route structures. As a use case, we will estimate the capacity, throughput, and maximum number of simultaneous operations in the route network in Dallas/Fort Worth (DFW) metropolitan area in Texas. As shown in Fig. 3, there are seven vertiports around the DFW urban area (labeled with white text each beginning with 'DF'), and these vertiports are connected by twelve routes. This route structure was defined considering the controlled airspace restrictions for conventional air traffic at major airports in this area like Dallas/Fort Worth international airport and Dallas Love Field airport, and it was used in the X4 simulations in which NASA and industry partners collaborated to evaluate strategic conflict management capabilities for UAM operations [11, 12].

A. Throughput analysis for a given DFW route network

Table 2 shows the estimated flight times of twelve origin-destination pair routes used in the X4 simulations, rounded up in minutes. Considering the crossing and merging waypoints between routes and the overlapping route segments shared by multiple routes, a set of exclusive routes are also selected. To maximize the number of simultaneous operations, the routes having longer flight times are chosen.

In the X4 simulations, the capacity at all vertiports was assumed to be 4 operations per a 12-minute time bin for demand and capacity balancing, which corresponds to $C_{vpt} = 20$ operations/hour. In this study, we also assumed that the minimum separation time is 2 min, N_f is 40 aircraft, $T_{operation}$ is 25 min (sum of average flight time of 15 min and turnaround time of 10 min), the maximum angle at crossing waypoints is 120 degrees, and C_{sector} is still a large number M . Then, the capacity of this route network is computed as follows:

$$Capacity_{Network} = \min \left\{ 20 * \frac{7}{2}, \frac{60}{2} * 11, \frac{60}{2 * 2} * 10 + \frac{60}{2}, M, \frac{40}{25/60} \right\} = \min \{ 70, 330, 180, M, 96 \} = 70 (ops/hr) \quad (13)$$

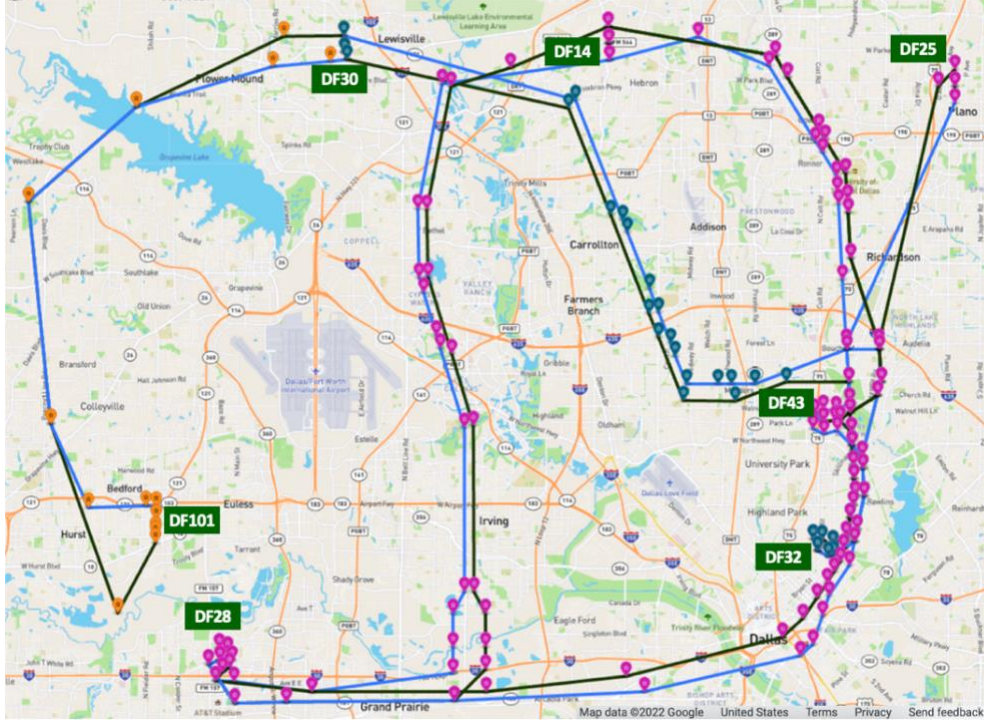


Fig. 3 Route network in Dallas/Fort Worth urban area for X4 Simulation.

Note that the route between DF30 and DF101 is bi-directional, reducing the number of effective routes to 11 and lowering the in-trail route capacity. This route is also independent in the given network and doesn't have any crossing waypoint with other routes. So, its capacity value is separately added to the crossing waypoint capacity for the other ten routes having crossing or merging points. In this use case, we can see that the most significant factor to limit the capacity of this route network is the vertiports. To increase the total network capacity and accommodate more traffic, the vertiport capacity should be increased by either adding more vertipads (or aircraft parking spaces) or shortening the operation time for takeoff and landing. The network capacity may also be increased by adding more vertiports, which will increase the route capacity as well.

Table 2: Flight times (in minutes) for origin-destination pair routes in DFW urban area.

No	Origin	Destination	Flight Time	Exclusive Routes
1	DF25	DF32	10	
2	DF32	DF25	10	
3	DF32	DF30	18	
4	DF30	DF32	17	
5	DF30	DF101	17	X
6	DF101	DF30	16	
7	DF14	DF43	11	X
8	DF14	DF28	20	X
9	DF43	DF14	13	X
10	DF43	DF28	15	
11	DF28	DF14	20	X
12	DF28	DF43	16	

In addition, the maximum number of simultaneous operations that this route network can achieve during a peak period is calculated below:

$$N_{ops} = \sum_{j \in R'} \frac{FT_j}{T_{sep}} = \frac{17+11+20+13+20}{2} = \frac{81}{2} = 40.5 \text{ aircraft} \quad (14)$$

However, note that the available fleet size in this use case is assumed to be 40 aircraft. So, the actual number of simultaneous operations that the UAM operators can serve in this route network will be limited up to 40 aircraft.

In traditional airline operations, the actual traffic demand usually has peaks and valleys appearing over time. It is expected that UAM operations will have a similar traffic pattern for its on-demand air transportation needs. Therefore, for the realistic traffic scenarios having peaks that were used in the X4 simulations, the practical throughput as a function of time can also be calculated, considering the spillover demand and the network capacity using Eqs. (7), (8), and (10). Figure 4 shows the traffic demand having several peaks in a 90-min long traffic scenario used for the X4 Scenario 1 Collaborative Simulation [11, 12], the major capacities for the given DFW route structure, and the practical throughput over time (a time unit for discretization is 3 min). In this normal operation condition, all the capacity values are assumed to be constant over time. The analysis result shows that the actual throughput is determined by the demand when the traffic level is low, but the throughput is limited by the vertiport capacity when the traffic demand is high. Due to the imbalance between demand and capacity, the excess demand is shifted to the later time, causing delay.

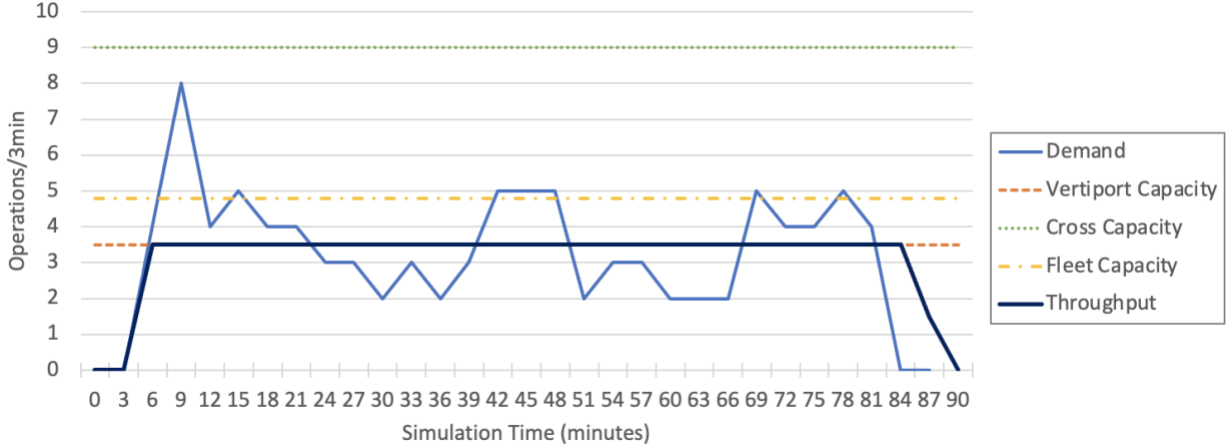


Fig. 4 Traffic demand, capacity, and practical throughput profiles for a normal traffic scenario.

B. Off-nominal scenario 1: Airspace restriction

If an off-nominal event occurs during the UAM operations, the capacity may change, affecting the throughput. In this subsection, we consider a temporary flight restriction in the airspace as a typical off-nominal scenario. In fact, this scenario was explored in the X4 Scenario 2 Collaborative Simulation to evaluate the re-routing capability for airborne or pre-departure flights in case of a UAS Volume Reservation (UVR) pop-up, which would be announced for public safety or some disasters such as fires in the area on the route of the planned operation [11]. In the X4 simulation, the UVR was announced in the middle of the simulation and required the UAM operators to modify their flight plans to reroute around the restricted airspace or assign pre-departure delay. The UVR was located over the course of the routes between DF14 and DF43, so the flights using these routes while the UVR is activated would be affected.

From the capacity perspective, this airspace restriction affects the in-trail route capacity only. When the UVR is activated, two routes between DF14 and DF43 are blocked. Therefore, the total in-trail route capacity is reduced to $60 / 2 * 9 = 270$ operations/hour. However, this capacity value is still higher than other terms like vertiport capacity and fleet availability. So, the UVR pop-up doesn't change the total network capacity, and the throughput also remains the same.

With respect to the number of simultaneous operations, a new set of exclusive routes should be chosen. By replacing the routes between DF14 and DF43 with the routes between DF25 and DF32, the maximum number of simultaneous operations when the UVR exists can be estimated as follows:

$$N_{ops} = \sum_{j \in R'} \frac{FT_j}{T_{sep}} = \frac{17+10+20+10+20}{2} = \frac{77}{2} = 38.5 \text{ aircraft} \quad (14)$$

In actual operations, however, several flights that already took off before the UVR is announced would keep flying with a detour route around the restricted area. So, the actual number of concurrent flights can be slightly higher than the estimation.

C. Off-nominal scenario 2: Vertiport closure

Another off-nominal event we can consider is a vertiport closure. A specific vertiport can be closed anytime for some reasons such as fires, emergency landing, and facility problems. In case of a vertiport closure, it is assumed that all the takeoffs or landings are prohibited, which makes an impact on the vertiport capacity directly. In addition, as the origin-destination pair routes are dependent on the vertiport, it also reduces the route capacity by the number of routes connected to this closed vertiport. UAM vehicles parked on the vertiport for the next flight would not be available for the transportation service. Therefore, the vertiport closure can influence all the capacity terms considered in this study. Table 3 shows how the capacity values change when each vertiport is closed in the given route network for DFW metropolitan area. In this case, the vertiport is still a main factor that determines the total network capacity.

Table 3: Capacity reduction due to a specific vertiport closure.

Closed Vertiport	Vertiport Capacity	In-trail Route Capacity	Crossing Waypoint Capacity	Fleet Availability Capacity	Network Capacity
None	70	330	180	96	70
DF25	60	270	150	80	60
DF32	60	210	120	64	60
DF30	60	210	120	64	60
DF101	60	270	150	80	60
DF14	60	210	120	64	60
DF28	60	210	120	64	60
DF43	60	210	120	64	60

Figure 5 illustrates the traffic demand, capacity, and resultant throughput profiles in the case that vertiport DF43 is temporarily closed from 30min to 60min after simulation starts. During the vertiport closure time, it can be observed that the capacity lines are lowered and that the practical throughput is limited by the reduced vertiport capacity. Compared to the normal scenario shown in Fig. 4, as a result, more flights experience delays up to the end of simulation by the amount of the increased gap between demand and capacity.

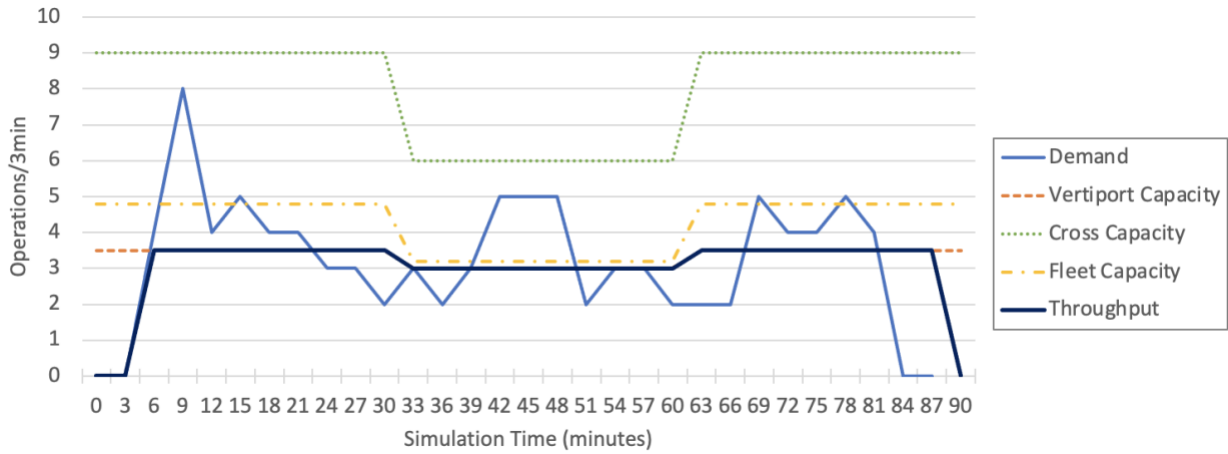


Fig. 5 Traffic demand, capacity, and practical throughput profiles for a vertiport closure scenario.

VII. Conclusions

In this paper, we considered various factors that could affect the capacity of airspace resources needed for UAM operations and developed an analytical model to estimate the capacity and throughput of the given UAM route network. The proposed throughput estimation model was evaluated with the initial UAM route structure in Dallas/Fort Worth area at low operational tempo and considered off-nominal scenarios as well. The proposed method enables estimation of the achievable throughput, as well as the maximum number of simultaneous operations, with respect to a new airspace construct design for UAM operations without any complex simulation modeling effort. It also allows us to identify which resource is the main bottleneck to limit the total network capacity, which may be useful to come up with recommendations on infrastructure investment, airspace construct design, and procedure changes for UAM operational environments. As shown in Section VI, this model can predict how the practical throughput would vary as demand or capacity changes over time.

This UAM throughput estimation model can be applied to different route networks such as more complex route structures for higher operational tempo and new UAM route networks in other cities having different operational conditions. For future work, this model will be improved by accounting for additional factors contributing to the capacity such as noise restriction, flight rules, and human resource availability like pilots and controllers, as well as by considering uncertainties and wind impacts. In addition, the throughput metrics roughly estimated by this analytical model will be compared with the more accurate values obtained by high-fidelity fast-time simulations like Autonomy Development toolkit (ADK) for various traffic scenarios [23].

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